Connectome data: APL synapse tables and per-cell counts

> **QUESTION**
> Where do the APL synapse tables come from, what does one row mean, how are synapses counted, and how many synapses does APL make with each kind of partner in each part of the mushroom body?

INPUTS

- Public synapse tables of four adult fly connectomes (downloaded 2026-10-07; sources in Table 2)

OUTPUTS

- All in `with_outside_ROI_synapses/` (the corrected version used from 2026-11-03; the official-ROI files of the same names in the main folder are unchanged)
- `<dataset>_APL_MB_synapses.csv.gz`: one row per APL synapse in the mushroom body (four datasets)
- `KC_weights_APL_PN_per_KC.csv`: per Kenyon cell, its PN input and its synapses with APL (calyx and whole mushroom body)
- `PN_calyx_weights_APL_vs_KC.csv`, `PN_MB_weights_APL_vs_KC.csv`: per projection neuron, its synapses onto KCs and with APL
- `summary_counts_by_compartment_and_partner_class.csv`, the scatter plots `scatter*.png`

SCRIPTS

- `scripts/hemibrain.py`, `malecns.py`, `malecns2.py`, `fafb_princeton.py`, `banc_extract.py` (download and filter)
- `scripts/prep.py`, `run.py`, `export.py` (regions, compartments, claws, boutons, export)
- `scripts/scatter.py`, `mb_plots.py`, `pnmb_*.py` (per-cell counts and scatter plots)
- `scripts/reports/r01_connectome_tables.py` (this report)

← Report 00 · All reports · Report 02 →

CONTENTS

1 The four connectomes and how each synapse is counted
2 What one row of the table is
3 Regions of the mushroom body and how many APL synapses each has
  3.1 Who APL talks to
4 Lobe compartments
5 Per-cell counts: Kenyon cells and projection neurons
6 Caveats

Every later step starts from the tables described here. Reports 02 and 03 add structure to them (claws, boutons, odor responses); Reports 04 to 11 use the hemibrain and male CNS parts of them to build and test a model.

Which version. Since 2026-11-03 the reports use the corrected tables: every synapse inside the official mushroom body regions of interest (ROIs), plus synapses just outside them that belong to the same calyx, peduncle or lobe (within 3 µm; Report 02 gives the rule). The male CNS calyx ROI misses the back of the calyx, so this matters most there. The added rows are 18.1% (FlyWire 783), 1.6% (Hemibrain v1.2), 11.8% (Male CNS v1.0), 18.8% (BANC 888) of each table. The official-ROI-only tables are kept unchanged next to them, and the earlier version of this report used them.

Table 1. Symbols used in this report.

Symbol	What it stands for	Values, units, range

Symbol	What it stands for	Values, units, range
$i,\ j$	Two neurons; i presynaptic, j postsynaptic	neuron IDs of each dataset
w_{ij}	Synaptic weight from i to j: the number of synaptic connections (T-bar → postsynaptic density pairs) from i onto j, counted inside the region named	whole number ≥ 0; same as neuPrint "weight" and FlyWire "syn_count"
$\mathcal{R}$	A region of the mushroom body: calyx (CA), peduncle (PED) or a lobe	official ROI of each dataset
$n_k^{\mathrm{PN}\to\mathrm{KC}}$	PN input of Kenyon cell k: $\sum_{i\in\mathrm{PNs}} w_{ik}$ in the calyx	synapses per KC
$n_k^{\mathrm{KC}\to\mathrm{APL}}$	Synapses from KC k onto APL, $w_{k,\mathrm{APL}}$, in the calyx (or the whole mushroom body where stated)	synapses per KC
$n_k^{\mathrm{APL}\to\mathrm{KC}}$	Synapses from APL onto KC k, $w_{\mathrm{APL},k}$, in the calyx (or the whole mushroom body)	synapses per KC
$n_i^{\mathrm{PN}\to\mathrm{APL}},\ n_i^{\mathrm{APL}\to\mathrm{PN}}$	Synapses between projection neuron i and APL in the calyx, each direction	synapses per PN
ρ_s	Spearman rank correlation over cells	−1 to 1
k_{NN}	Number of nearest dopaminergic T-bars that vote on a lobe synapse's compartment (Section 4)	15

1 The four connectomes and how each synapse is counted

A connectome is a map of every neuron and synapse in a piece of nervous tissue, traced from electron microscopy. A chemical synapse is seen as one presynaptic release site (the **T-bar**) facing one or more postsynaptic densities (**PSDs**) on other neurons. All tables here count **T-bar → PSD pairs**: one T-bar facing three PSDs on three cells counts once for each cell, so the weight w_{ij} is the number of PSDs on j that face T-bars of i. This is the convention of neuPrint (hemibrain, male CNS) and FlyWire.

$$w_{ij}(\mathcal{R}) \;=\; \#\{(t,s)\ :\ t \text{ is a T-bar of } i,\ s \text{ is a PSD of } j \text{ facing } t,\ s\in\mathcal{R}\} \tag{1}$$

A synapse belongs to region $\mathcal{R}$ by the position of its postsynaptic site (hemibrain, male CNS) or by the dataset's own neuropil label (FlyWire, BANC).

The APL neuron (anterior paired lateral) is one large GABAergic neuron per brain hemisphere. It covers the whole mushroom body, receives input from the Kenyon cells and inhibits them back. Every APL synapse inside a mushroom body ROI, or just outside it (see above), was kept, in both directions and with every partner.

Table 2. The four connectomes. Rows = APL synapses in the mushroom body, corrected version; in brackets, how many rows the synapses just outside the official ROIs add. Every source is a public, no-login download. Hemibrain and male CNS come from the FlyEM team (Janelia), FlyWire from the FlyWire consortium (Princeton), BANC from the BANC consortium. The model in Reports 05 to 11 uses only hemibrain and male CNS (Yang, 2026-10-21).

Dataset	Animal	Source table	Filter	APLs	Rows
FlyWire 783 (FAFB)	female, whole brain	Codex "Princeton" synapse table, v783	proofread neurons, no autapses, ≥5 synapses per neuron pair (Codex default)	2 (left, right)	327,490 (+18.1%)

Dataset	Animal	Source table	Filter	APLs	Rows
Hemibrain v1.2	female, central brain, right side	neuPrint release feather files (synapse partners and points)	release confidence thresholds	1 (APL_R; the left APL is outside the volume)	245,662 (+1.6%)
Male CNS v1.0	male, brain and nerve cord	neuPrint release, `min conf-0.5` partner table	synapse confidence ≥ 0.5	2 (APL_L, APL_R)	491,427 (+11.8%)
BANC 888	female, brain and nerve cord	BANC v2 enriched synapse table	synapse size ≥ 5 (as in the BANC paper)	2 (left, right)	79,896 (+18.8%)

Why BANC counts are low

BANC has 3–5 times fewer APL synapses than the other datasets (median KC↔APL synapses per KC in the calyx: 2 vs 10 in hemibrain). This follows from its synapse detector and the size ≥ 5 filter, and it is why BANC numbers are treated as rough throughout.

Two FlyWire versions

The FlyWire rows in the main table keep only neuron pairs with ≥5 synapses (Codex default). The per-KC and per-PN counts (Section 5) instead use every Princeton synapse between proofread neurons, without that threshold, because a KC often makes fewer than 5 synapses with APL in the calyx. So FlyWire per-cell counts are larger than counts from the main table.

2 What one row of the table is

Each row is one T-bar → PSD pair where APL is on one side. The columns that the later steps use:

Table 3. Main columns of `<dataset>_APL_MB_synapses.csv.gz`. The README lists every column.

Column	Meaning
`direction`	`APL_output (APL presynaptic)`: APL → partner. `APL_input (APL postsynaptic)`: partner → APL.
`apl_id`, `apl_side`	Which APL. Each mushroom body has its own APL, so "APL side" = which mushroom body.
`partner_id`, `partner_type`, `partner_class`	The other neuron, its cell type and its broad class (blank = untyped fragment)
`pre_*_nm`, `post_*_nm`	Positions of the T-bar and the PSD, in nanometres, in the dataset's own coordinate space
`partner_side_*_nm`	The partner's own site (the PSD when APL is presynaptic, the T-bar when APL is postsynaptic); used to place a synapse on a claw or bouton
`roi_official`, `mb_region`	The dataset's own ROI label, and the mushroom body region: CA, PED, or a lobe
`mb_compartment`, `mb_compartment_source`	The finest compartment label (α1 … γ5 in the lobes) and whether it is official or derived (Section 4)

Column	Meaning
`claw_id`, `bouton_id` (and `nearest_*`, `*_dist_nm`)	The KC claw or PN bouton the synapse sits on, if any (Report 02)

3 Regions of the mushroom body and how many APL synapses each has

The mushroom body has three parts. The **calyx** (CA) is where projection neurons (PNs) from the antennal lobe pass odor information to the dendritic claws of the Kenyon cells (KCs). The **peduncle** (PED) is the bundle of KC axons leaving the calyx. The **lobes** are where KC axons end and talk to output neurons (MBONs) and dopaminergic neurons (DANs). Hemibrain and male CNS split the lobes into α, α′, β, β′ and γ lobes; FlyWire and BANC only into a vertical lobe (VL: α + α′) and a medial lobe (ML: β + β′ + γ).

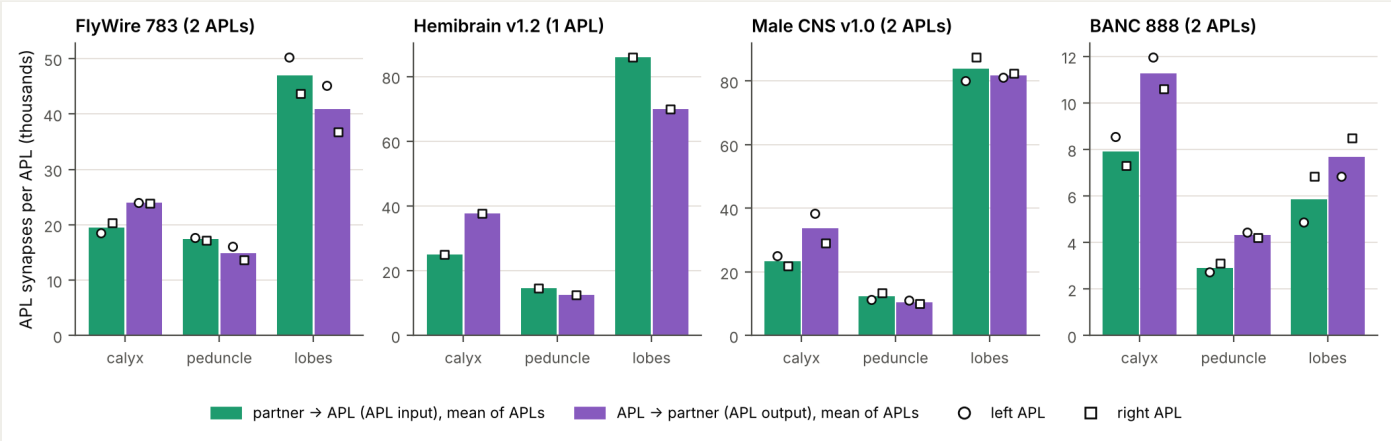


Figure 1. APL synapses per region and direction, with each APL shown. Each panel is one connectome. Bars count the rows of `<dataset>_APL_MB_synapses.csv.gz` (corrected version) in each region, averaged over the dataset's APLs, so the height is the synapses of one average APL. The white markers on each bar are the single APLs, one per hemisphere: circle = left APL, square = right APL. Hemibrain has only the right APL, so its marker sits on the bar. Green = synapses onto APL (APL is postsynaptic), purple = synapses made by APL (APL is presynaptic). Lobes = all lobe regions together. y-axis: thousands of T-bar → PSD pairs. The y-axes differ between panels: BANC has far fewer synapses (Section 1).

In every dataset APL makes more synapses than it receives in the calyx (APL output share of calyx APL synapses: FlyWire 783 55%, Hemibrain v1.2 60%, Male CNS v1.0 59%, BANC 888 59%). In the lobes the two directions are closer to equal.

3.1 Who APL talks to

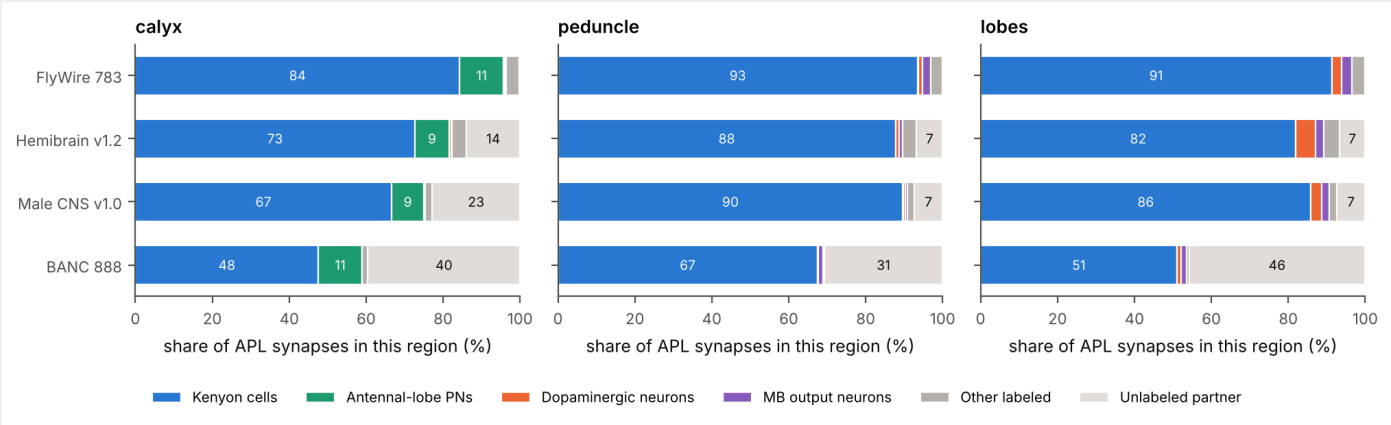


Figure 2. Partner classes of APL synapses, per region. For each region (panel) and dataset (row), the bar splits all APL synapses there (both directions) by the partner's class: Kenyon cells, antennal-lobe projection neurons (PNs), dopaminergic neurons (DANs), mushroom body output neurons (MBONs), other neurons with a class label, and unlabeled partners (untyped fragments). Numbers inside bars are percentages (shown when above 6%). The class labels come from each dataset's own annotation: hemibrain from the BANC team's compiled table, the others from their own releases. BANC has many unlabeled partners (39% of rows).

Kenyon cells are by far APL's main partner everywhere. In the calyx, PNs are the second partner (9–12% of APL synapses there): APL inhibits the PN boutons as well as the KC claws, which Reports 04 and 05 use.

---

4 Lobe compartments

Each lobe is divided into compartments (α1–3, α′1–3, β1–2, β′1–2, γ1–5), each defined by which DAN and MBON types innervate it. Only the male CNS publishes compartment volumes (`malecns-subcompartments-v3`); there the official label is used. For the other datasets the compartment of a lobe synapse is **derived** from the dopaminergic neurons around it:

1. In male CNS, measure for each DAN family (e.g. PAM12, PPL101) what fraction of its T-bars falls in each compartment: $P(c \mid \phi)$ for compartment c and family ϕ.
2. For a synapse s in a lobe, take the $k_{\mathrm{NN}} = 15$ nearest DAN T-bars inside the same lobe ROI.
3. Each of them votes with its family's distribution. The compartment with the largest summed vote wins (Equation 2); `compartment_vote_share` is that vote divided by the total, a confidence between 0 and 1.
4. A synapse whose nearest DAN T-bar is more than 8 µm away gets no compartment.

$$\hat{c}(s) \;=\; \arg\max_{c \,\in\, \mathrm{lobe}(s)} \sum_{d \,\in\, \mathrm{kNN}(s)} P\big(c \mid \phi(d)\big) \tag{2}$$

$\hat{c}(s)$: derived compartment of synapse s; $\mathrm{kNN}(s)$: its 15 nearest DAN T-bars in the same lobe; $\phi(d)$: the family of DAN T-bar d.

Check against the official volumes. Applied to the male CNS itself, the derived label matches the official one for 96.9% of 311,252 lobe synapses when the lobe is known (the hemibrain situation). The weakest compartment is γ3 (78%). With only the coarse VL/ML lobes (FlyWire, BANC) the agreement was 88%, mainly because α′2 is then called α2. For the per-KC lobe plots a refined vote (DAN T-bars restricted to the KC's own lobe) reaches 97.9%.

Not used by the model

Compartments matter only for the lobe scatter plots (`scatter4` – `scatter7`). The model in Reports 05–11 uses calyx synapses only.

5 Per-cell counts: Kenyon cells and projection neurons

Two tables sum the synapses per cell. `KC_weights_APL_PN_per_KC.csv` has one row per KC with its calyx PN input $n_k^{\mathrm{PN}\to\mathrm{KC}}$, its calyx synapses onto APL $n_k^{\mathrm{KC}\to\mathrm{APL}}$ and from APL $n_k^{\mathrm{APL}\to\mathrm{KC}}$, and the same three counts over the whole mushroom body (`*_MB` columns). KC class (αβ, α′β′, γ) comes from the KC type name. `PN_calyx_weights_APL_vs_KC.csv` has one row per antennal-lobe PN with its calyx synapses onto KCs and to and from APL.

$$n_k^{\mathrm{PN}\to\mathrm{KC}} = \sum_{i\,\in\,\mathrm{PNs}} w_{ik}(\mathrm{CA}), \qquad n_k^{\mathrm{KC}\to\mathrm{APL}} = w_{k,\mathrm{APL}}(\mathrm{CA}), \qquad n_k^{\mathrm{APL}\to\mathrm{KC}} = w_{\mathrm{APL},k}(\mathrm{CA}) \tag{3}$$

Table 4. Kenyon cells per class. Counted from `KC_weights_APL_PN_per_KC.csv` (KCs with at least one PN or APL synapse in the mushroom body). FlyWire, male CNS and BANC have both hemispheres; hemibrain has one. FlyWire has about twice as many γ KCs per side as hemibrain (a known difference, also reported by Schlegel et al. 2024). "No subtype" = BANC KCs without a class in their type name.

Dataset	αβ	α′β′	γ	no subtype	all KCs
FlyWire 783	1,772	916	2,489	0	5,177
Hemibrain v1.2	889	340	708	0	1,937
Male CNS v1.0	1,810	695	1,556	2	4,063
BANC 888	1,452	605	2,036	380	4,473

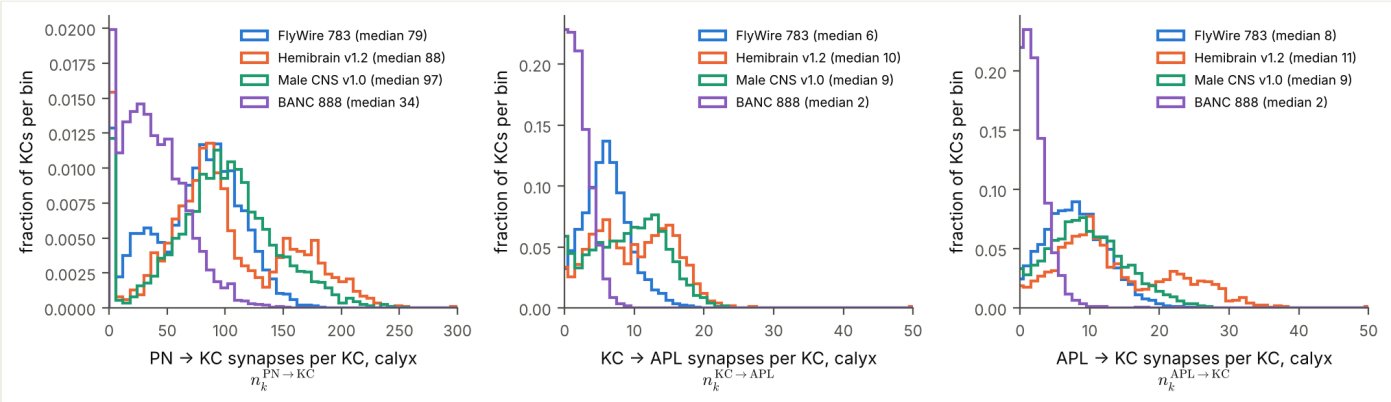


Figure 3. How many PN and APL synapses a Kenyon cell has in the calyx. Distributions over all KCs of a dataset (both hemispheres where present) of the three calyx counts defined in Equation 3. Each line is one connectome (colors in the legend, with the median). x-axis: synapses per KC (T-bar → PSD pairs); values above the axis end (300 or 50) are piled into the last bin. y-axis: fraction of the dataset's KCs per bin (bin width 6 synapses in the first panel, 1 in the others). The peaks at 0 are mostly γd and αβp KCs, whose dendrites sit in the accessory calyces outside the calyx ROI: they have no PN input and no APL synapse in the main calyx (6% of KCs in FlyWire 783, 7% of KCs in Hemibrain v1.2, 7% of KCs in Male CNS v1.0, 6% of KCs in BANC 888).

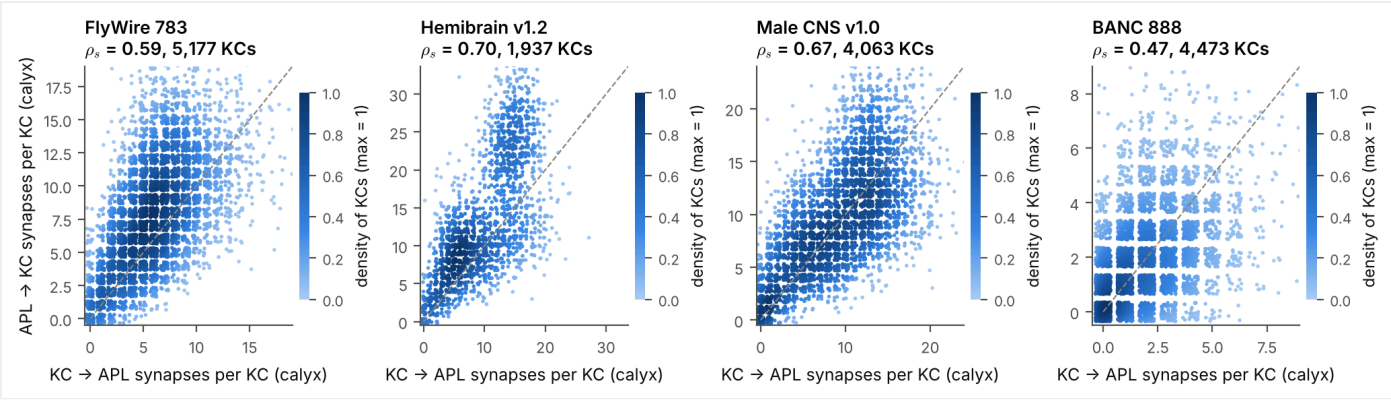


Figure 4. Reciprocity between KCs and APL in the calyx. Each panel is one connectome; each dot is one KC, placed by its calyx synapses onto APL (x) and from APL (y). The counts are whole numbers, so each dot is shifted by a random amount of up to ±0.35 to keep KCs with the same counts from hiding each other. Color shows how crowded that part of the plot is (Gaussian kernel density of the dots, scaled so the densest point is 1; dark = many KCs). Dashed line: equal counts. Axes end at the 99.5th percentile. ρ_s = Spearman rank correlation over KCs. KCs that give APL many synapses also receive many from it: the connection is reciprocal at the level of single cells.

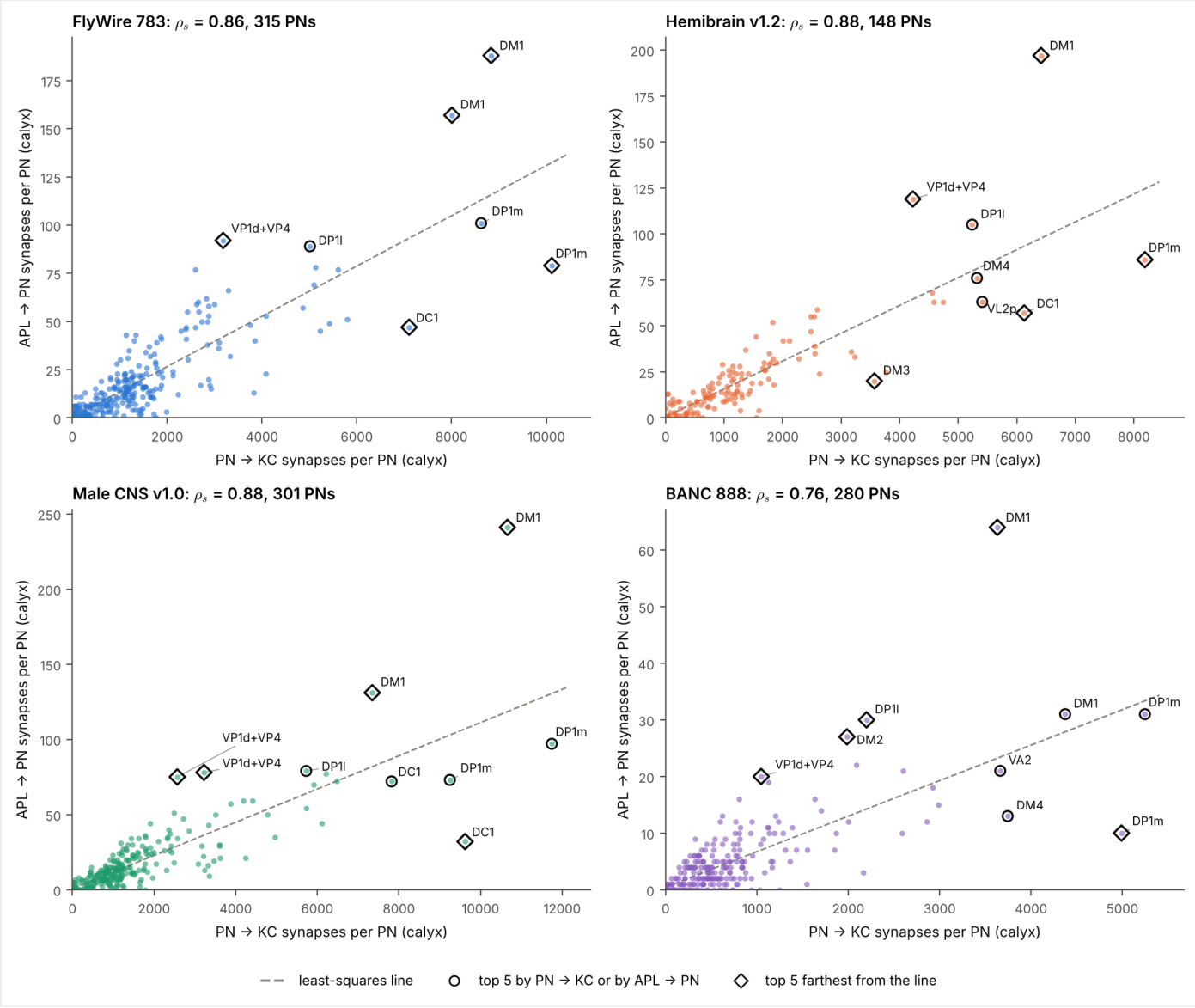


Figure 5. APL input to each projection neuron vs its output to Kenyon cells, calyx. Each dot is one antennal-lobe PN with at least one calyx synapse onto a KC. x: its synapses onto KCs in the calyx; y: synapses it receives from APL in the calyx. Dashed line: least-squares line over the PNs of that dataset. Labelled PNs (glomerulus, or the type name for multiglomerular PNs): circles are the 5 PNs with the most synapses onto KCs and the 5 with the most synapses from APL (often the same PNs); diamonds are the 5 PNs farthest above or below the line, i.e. with the most or least APL input for their output. Table 6 lists them with their numbers. ρ_s = Spearman rank correlation over PNs. PNs with more output to KCs (more or bigger boutons) get more APL inhibition, roughly in proportion. Report 04 uses the ratio of the two, per glomerulus, as the bouton weight p_g.

Table 5 gives the means behind these figures.

Table 5. Mean synapses per cell. First five columns: per KC (calyx unless "whole MB"); last three: per antennal-lobe PN, calyx. FlyWire counts here have no per-pair threshold (Section 1). Even with the synapses outside the calyx ROI added back, male CNS KC↔APL calyx counts per KC stay below hemibrain's (Report 02).

Dataset	PN→KC per KC	KC→APL per KC	APL→KC per KC	KC→APL per KC, whole MB	APL→KC per KC, whole MB	PN→KC per PN	PN→APL per PN	APL→PN per PN
FlyWire 783	73.0	6.1	8.0	28.7	28.2	1200	16	16
Hemibrain v1.2	96.1	9.9	13.5	52.4	49.3	1258	19	20

Dataset	PN→KC per KC	KC→APL per KC	APL→KC per KC	KC→APL per KC, whole MB	APL→KC per KC, whole MB	PN→KC per PN	PN→APL per PN	APL→PN per PN
Male CNS v1.0	96.1	9.1	9.7	51.2	47.6	1298	17	15
BANC 888	37.6	2.0	2.1	4.4	5.0	601	11	4

Table 6. The labelled PNs of Figure 5. "Most synapses": the union of the 5 PNs with the most PN→KC synapses and the 5 with the most APL→PN synapses, sorted by PN→KC. "Farthest from the line": the 5 PNs with the largest distance above (+) or below (−) the least-squares line, in APL→PN synapses. "Both": in both groups. Each row is one PN, sorted by PN→KC within a dataset. Calyx synapses, corrected version.

Dataset	group	label	PN type	PN→KC	APL→PN	APL→PN per 100 PN→KC	distance from the line (APL→PN)
FlyWire 783	both	DP1m	`DP1m_adPN`	10,114	79	0.8	−53
FlyWire 783	both	DM1	`DM1_lPN`	8,832	188	2.1	+72
FlyWire 783	most synapses	DP1m	`DP1m_adPN`	8,623	101	1.2	−12
FlyWire 783	both	DM1	`DM1_lPN`	8,007	157	2.0	+52
FlyWire 783	both	DC1	`DC1_adPN`	7,104	47	0.7	−46
FlyWire 783	most synapses	DP1l	`DP1l_adPN`	5,012	89	1.8	+23
FlyWire 783	both	VP1d+VP4	`VP1d+VP4_l2PN1`	3,178	92	2.9	+50
Hemibrain v1.2	both	DP1m	`DP1m_adPN`	8,188	86	1.1	−38
Hemibrain v1.2	both	DM1	`DM1_lPN`	6,413	197	3.1	+99
Hemibrain v1.2	both	DC1	`DC1_adPN`	6,126	57	0.9	−36
Hemibrain v1.2	most synapses	VL2p	`VL2p_adPN`	5,408	63	1.2	−19
Hemibrain v1.2	most synapses	DM4	`DM4_adPN`	5,318	76	1.4	−5
Hemibrain v1.2	most synapses	DP1l	`DP1l_adPN`	5,237	105	2.0	+25
Hemibrain v1.2	both	VP1d+VP4	`VP1d+VP4_l2PN1`	4,223	119	2.8	+55
Hemibrain v1.2	farthest from the line	DM3	`DM3_adPN`	3,566	20	0.6	−35
Male CNS v1.0	most synapses	DP1m	`DP1m_adPN`	11,746	97	0.8	−33

Dataset	group	label	PN type	PN→KC	APL→PN	APL→PN per 100 PN→KC	distance from the line (APL→PN)
Male CNS v1.0	both	DM1	DM1_lPN	10,660	241	2.3	+123
Male CNS v1.0	both	DC1	DC1_adPN	9,622	32	0.3	−75
Male CNS v1.0	most synapses	DP1m	DP1m_adPN	9,249	73	0.8	−30
Male CNS v1.0	most synapses	DC1	DC1_adPN	7,821	72	0.9	−15
Male CNS v1.0	both	DM1	DM1_lPN	7,347	131	1.8	+49
Male CNS v1.0	most synapses	DP1l	DP1l_adPN	5,727	79	1.4	+15
Male CNS v1.0	both	VP1d+VP4	VP1d+VP4_l2PN1	3,223	78	2.4	+42
Male CNS v1.0	farthest from the line	VP1d+VP4	VP1d+VP4_l2PN1	2,569	75	2.9	+46
BANC 888	most synapses	DP1m	DP1m_adPN	5,247	31	0.6	−2
BANC 888	both	DP1m	DP1m_adPN	4,990	10	0.2	−22
BANC 888	most synapses	DM1	DM1_lPN	4,374	31	0.7	+3
BANC 888	most synapses	DM4	DM4_adPN	3,744	13	0.3	−11
BANC 888	most synapses	VA2	VA2_adPN	3,663	21	0.6	−2
BANC 888	both	DM1	DM1_lPN	3,631	64	1.8	+41
BANC 888	both	DP1l	DP1l_adPN	2,199	30	1.4	+16
BANC 888	both	DM2	DM2_lPN	1,985	27	1.4	+14
BANC 888	farthest from the line	VP1d+VP4	VP1d+VP4_l2PN1	1,046	20	1.9	+13

6 Caveats

- **Detection differs between datasets.** Each connectome has its own synapse detector and thresholds, so absolute counts are not comparable across datasets; ratios and ranks within a dataset are safer. The model (Reports 05–11) works with ratios and ranks.
- **Male CNS calyx ROI.** The official CA(L)/CA(R) ROIs leave out the posterior part of each calyx: 18% (left) and 9% (right) of PN↔APL and 33% / 14% of KC↔APL calyx synapses fall outside them. This report uses the corrected version, which adds them back (Report 02).
- **Untyped partners.** Partners without a type are 9% of hemibrain rows, 10% of male CNS rows and 39% of BANC rows.

- **Hemibrain types.** `partner_type` is the official neuPrint v1.2 type. A compiled table had labelled about 200 α′β′ KCs as γ; that was corrected on 2026-10-10.
- **Summary table.** `summary_counts_by_compartment_and_partner_class.csv` was made from an earlier hemibrain table; 5 hemibrain γ3–γ5 rows differ by 1–4 synapses from the main table.

← Report 00 · All reports · Report 02 →